

### 3 Curvature of spacetime

In this chapter we learn how spacetime is curved due to mass.

#### 3.1 The Equivalence Principle

According to Newton's second law of motion, the force applied to an object is proportional to its acceleration:

$$\vec{F} = m_i \vec{a}, \quad (6)$$

where the constant of proportionality,  $m_i$ , is called the inertial mass. On the other hand, the gravitational force acting on an object is proportional to the gravitational acceleration:

$$\vec{F}_g = m_g \vec{g}, \quad (7)$$

where  $m_g$  is called the gravitational mass. There is no fundamental reason to assume that inertial and gravitational mass are necessarily identical—they are, mathematically speaking, merely constants of proportionality. However, since the pioneering experiments by Eötvös, increasingly precise measurements have confirmed that the ratio  $m_i/m_g$  is equal to unity to a very high degree of accuracy. The statement that **inertial and gravitational mass are equal** is known as the **weak equivalence principle** (WEP) of Newtonian mechanics.

This principle has profound implications. Firstly, it implies that **one cannot distinguish between a uniform gravitational field and a uniformly accelerated reference frame**. To illustrate, consider an observer inside a sealed elevator on Earth's surface. By observing the motion of objects inside the elevator, the observer can determine the gravitational acceleration,  $g$ . Every experiment conducted in the elevator will lead to the conclusion that the observer is within a gravitational field. Now, suppose the elevator is inside a spaceship accelerating at  $g$  in empty space. If the observer repeats the same experiments, they will yield identical results. Consequently, the observer cannot distinguish between the effects of gravity and acceleration. Thus, the weak equivalence principle can also be stated as follows: **The motion of freely falling particles is identical in a uniform gravitational field and a uniformly accelerated frame, provided the region of spacetime under consideration is sufficiently small.** The qualification regarding a sufficiently small region is necessary because large-scale gravitational inhomogeneities could, in principle, be detectable.

This thought experiment suggests that gravity can be locally replaced by uniform acceleration. More generally, it implies that **one can eliminate the effects of gravity by adopting a coordinate system that is freely falling in a gravitational field**. However, this cancellation only works within a small enough region of space. For example, a freely falling elevator. Another example is the Earth itself is in free fall toward the Sun, which is why we do not directly feel the Sun's gravitational pull—it is locally canceled out. However, slight variations in the gravitational field due to the Sun (tidal effects) still produce observable consequences, such as ocean tides. **Thus, a freely falling frame is locally indistinguishable from an inertial frame.** How is a freely falling frame in a gravitational field equivalent to an inertial frame? Consider a laboratory floating in deep space, free from any gravitational influences—this is a truly inertial frame, meaning it is not accelerating. Now imagine the same laboratory falling freely toward the Earth. If an observer is trapped inside, they will find no way to distinguish the two scenarios through internal experiments. This fundamental insight leads to the **strong equivalence principle** (SEP), which generalizes the WEP: **In an arbitrary gravitational field, it is always possible to choose a locally inertial coordinate system such that, within a sufficiently small region of spacetime, the laws of physics take the same form as in an unaccelerated Cartesian coordinate system in the absence of gravity.** This principle is extremely useful because it allows one to locally construct an inertial frame even within a gravitational field, thereby enabling the application of special relativity in curved spacetime. In the statement above, the term "Cartesian" indicates that, in this locally inertial coordinate system, Newton's second law takes the form:

$$F = m \frac{d^2 x}{dt^2}. \quad (8)$$

It is straightforward to show how a freely falling frame eliminates gravitational effects. Consider a particle of mass  $m$  above the Earth's surface, subject to a force  $\vec{F}$ . In a Cartesian coordinate system, its equation of motion is:

$$m \frac{d^2 x}{dt^2} = m\vec{g} + \vec{F}. \quad (9)$$

Now, introduce a coordinate transformation to a freely falling frame:

$$x' = x - \frac{1}{2}gt^2. \quad (10)$$

Although one could also consider a transformation of time, for non-relativistic motion near Earth's surface, time remains the same for both the particle and an observer on Earth's surface. Substituting this transformation into the equation of motion yields:

$$m \frac{d^2 x'}{dt^2} = \vec{F}, \quad (11)$$

where the gravitational term has completely disappeared. This example illustrates that both the original observer using coordinates  $x$  and the freely falling observer using coordinates  $x'$  agree on the mechanics of the particle. The only difference is that the observer using  $x'$  does not experience gravity. To summarize:

- One cannot distinguish between a uniform gravitational field and uniform acceleration.
- One cannot differentiate an inertial frame from a frame that is freely falling under the influence of gravity alone.

## 3.2 Fermat's Principle

In a medium, light follows the path that takes the shortest time, not necessarily the shortest distance. This is known as Fermat's Principle, which states that the **path taken by light between two points is the one that minimizes (or extremizes) the travel time**. If the geometry in which the light travels is Euclidean or flat then the path is always a straight line. This principle is helpful to understand what happens to spacetime in the presence of mass.

## 3.3 Light and gravity: curvature of spacetime

An important consequence of the equivalence principle (EP) is that gravity curves spacetime. This can be understood through a thought experiment analyzing how light responds to a gravitational field. Naively, one might expect that light, being massless, is unaffected by gravity. However, consider the following experiment: you are inside a freely falling elevator moving towards Earth, while your friend remains stationary on the Earth's surface. You shine a laser from one side of the elevator to the other. Both of you trace the path of the light ray. Assuming the elevator is transparent, you both agree on where the light starts and where it hits the opposite wall.

**Your observation:** Since you are in an inertial frame and, according to the strong EP, the laws of physics in your frame are the same as in an inertial frame in the absence of gravity, you will observe the light traveling in a straight line.

**Your friend's observation:** By the time the light reaches the opposite wall, the elevator has moved slightly downward. Consequently, your friend sees the light follow a curved path, bending toward the Earth. Since both of you agree on the start and end points of the light's journey, this bending must be real. For further explanations, refer to [this resource](#). A similar experiment can be conducted in an accelerated frame: consider an elevator accelerating upward in empty space, as illustrated in Figure [11](#). If a beam of light enters from the outside and propagates towards the opposite wall, an observer inside the elevator will perceive the light bending downward. However, according to the EP, this scenario is equivalent to an elevator in a uniform gravitational field. Thus, we conclude that gravity must also bend light. In General Relativity (GR), the bending of light is interpreted as a consequence of spacetime curvature. Light follows a trajectory that minimizes travel time—these paths are called geodesics. In flat spacetime, geodesics are

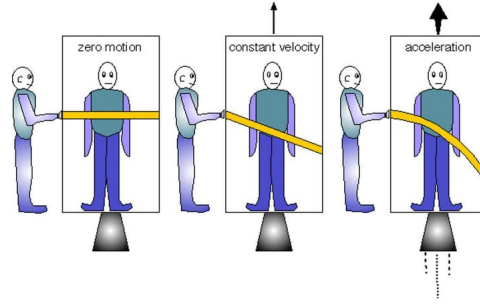


Figure 11: Light bends in a uniformly accelerated frame. According to the EP, light must also bend in a gravitational field.

straight lines, whereas, on a curved surface such as a sphere, they correspond to arcs of great circles. **Thus, the bending of light under gravity implies that the geometry that light follows is no longer Euclidean. This is equivalent of saying the spacetime is curved under gravity. Einstein's field equation in general relativity shows us how the presence of mass and energy curves the spacetime. According to this theory, gravity is simply the manifestation of the curvature of spacetime.**

### 3.4 Gravitational redshift

An important consequence of the equivalence principle (EP) is the phenomenon of *gravitational redshift*, which has been experimentally confirmed. **Gravitational redshift refers to the fact that when a light ray travels from a region of lower gravitational potential to a region of higher gravitational potential, its frequency is redshifted.** Consider a light source  $E$  at the surface of the Earth emitting light waves at a constant frequency, and a receiver  $R$  positioned at a height  $h$ . According to the equivalence principle, this setup is equivalent to both the emitter  $E$  and the receiver  $R$  accelerating with a constant acceleration  $g$ , maintaining a fixed distance of separation  $h$ . At time  $t_0$ , the emitter sends out a photon with frequency  $\nu_E$ . This photon reaches the receiver at time  $t = h/c$ , where  $c$  is the speed of light. During this interval, the receiver has acquired a velocity  $v = gt = gh/c$ . Using the classical Doppler effect, the received frequency  $\nu_R$  is given by:

$$\nu_R = \nu_E \left(1 - \frac{v}{c}\right) = \nu_E \left(1 - \frac{gh}{c^2}\right), \quad \frac{\nu_R - \nu_E}{\nu_E} = -\frac{gh}{c^2}. \quad (12)$$

This result shows that  $\nu_R < \nu_E$ , meaning the received light is redshifted. Fundamentally, gravitational redshift is a direct consequence of gravitational time dilation—clocks at different heights in a gravitational field tick at different rates. This effect is crucial in practical applications such as the calibration of the Global Positioning System (GPS). For further reading, see [this resource](#) or [this NASA article](#).

### 3.5 Metric in flat space and on the surface of a sphere

In the flat space

$$ds^2 = dx^2 + dy^2,$$

hence the metric is

$$\eta_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (13)$$

On the surface of a sphere of radius  $R$

$$ds^2 = R^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

and hence the metric is

$$\eta_{ij} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix} \quad (14)$$

### 3.6 FRW metric

By the statement "*the universe is homogeneous and isotropic*," we mean that at any given instant of time, the universe appears the same everywhere (homogeneity), and all spatial directions are equivalent (isotropy). However, homogeneity and isotropy are **relative properties** since they depend on the observer's frame of reference. As the real universe is filled with matter and radiation, only those observers who are at rest with respect to this matter and radiation perceive the universe as homogeneous and isotropic. Such observers are called *comoving observers*, and their reference frame is called the *comoving frame*. We will write the metric of the universe in a comoving frame.

**To write the metric of the universe in a comoving frame, we must recognize that homogeneity and isotropy are preserved over time as the universe expands.** Therefore, we can consider three-dimensional spatial hypersurfaces embedded in four-dimensional spacetime, which must be homogeneous and isotropic at each moment in time. The meaning of homogeneity and isotropy in these three-dimensional spatial hypersurfaces is that the metric describing space is invariant under **isometries** (transformations that preserve distances and angles).

These conditions imply that, in three dimensions, there are only three possible spatial geometries that satisfy homogeneity and isotropy:

1. **Flat space** (Euclidean, zero curvature),
2. **Three-dimensional sphere** (constant positive curvature),
3. **Three-dimensional hyperboloid** (constant negative curvature).

Three-dimensional hyper-surfaces are hard to visualize. **So we consider a two dimensional surface that is embedded in three dimensional hypersurface, for example a two-sphere (or sphere).** Generalization to three-dimensional hypersurfaces, embedded in four-dimension is then straight-forward. In Cartesian coordinate, the metric for 3D Euclidian space is  $ds^2 = (dx)^2 + (dy)^2 + (dz)^2$ . Therefore, the embedding of a two-sphere in three-dimensional Euclidian space is

$$x^2 + y^2 + z^2 = R^2,$$

where  $R$  is the radius of the sphere<sup>[1]</sup>. This equation represents constraints among the coordinates which should be incorporated in the metric. Differentiating the sphere we get

$$dz = -\frac{xdx + ydy}{z} = -\frac{xdx + ydy}{\sqrt{R^2 - x^2 - y^2}}.$$

Therefore, we can re-write the metric as

$$ds^2 = (dx)^2 + (dy)^2 + \frac{(xdx + ydy)^2}{R^2 - x^2 - y^2} \quad (15)$$

So the distance between two points on the surface of the sphere is given in terms of two independent coordinates. It is convenient to introduce angular coefficients  $(r', \theta)$  defined as

$$x = r' \cos \theta, \quad y = r' \sin \theta.$$

Substituting in the expression of  $ds^2$  above, we get the metric as

$$ds^2 = \frac{dr'^2}{1 - r'^2/R^2} + r'^2 d\theta^2.$$

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<sup>1</sup>One may think that the choice of a sphere is special. You can chose arbitrary shape  $ax^2 + by^2 + cz^2 = R^2$  where  $a, b, c$  are constant. But the constants can always be absorbed by the redefinition of the coordinates.

The  $0 < R^2 < \infty$  corresponds to a sphere, and the limit  $R^2 \rightarrow \infty$  corresponds to a flat surface. These surfaces are formally called surfaces of uniform positive curvature, and zero curvature, respectively. One can also have uniform negative curvature for  $R^2 < 0$  which is called a Lobachevski surface, or a hyperbolic surface. The sphere and the hyperbolic surfaces are non Euclidean, see figure [12](#).

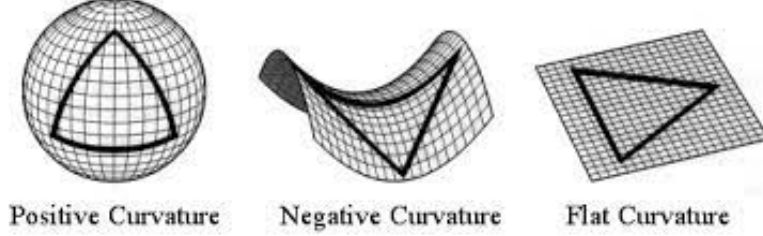


Figure 12: Geometry of two-space.

We can now obtain the metric on a three-surface. Introducing a fourth orthogonal direction  $\omega$ , the metric is

$$ds^2 = (dx)^2 + (dy)^2 + (dz)^2 + (d\omega)^2.$$

Now, consider a three-hypersurface of homogeneity and isotropy of radius  $R$

$$R^2 = x^2 + y^2 + z^2 + \omega^2.$$

Differentiation we get

$$d\omega = -\frac{xdx + ydy + zdz}{\sqrt{R^2 - x^2 - y^2 - z^2}}.$$

We now chose a set of coordinate  $(r, \theta, \phi)$  such that

$$x = r' \sin \theta \cos \phi, \quad y = r' \sin \theta \sin \phi, \quad z = r' \cos \theta.$$

This leads us to the expression of a metric on three hypersurface of homogeneity and isotropy as

$$ds^2 = \frac{dr'^2}{1 - r'^2/R^2} + r'^2(d\theta^2 + \sin^2 \theta d\phi^2).$$

Introducing a re-scaled coordinate  $r = r'/\sqrt{|R^2|}$ , we get

$$ds^2 = R^2 \left( \frac{dr^2}{1 - Kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right),$$

where we have introduced  $K$  which can be  $\pm 1$  or  $0$ . This is to avoid making the  $R^2$  negative. This is the metric for a homogeneous and isotropic three-surface. This serves as the spatial part of the metric of the four-dimensional universe.

We now embed this three-surface in the four-dimensional spacetime. One of the ingredient needed is the time  $t$ . This has to be introduced in Lorentzian sense. As the spatial part of the metric given above scales with time, the homogeneity and isotropy imply that the scaling must be same for the  $r, \theta, \phi$ . Without any loss of generality we replace  $R^2$  by a time-dependent factor,  $a(t)$  which is called *scale factor*. So the desired metric, called the FRW metric, for homogeneous, isotropic, and expanding universe is given by

$$ds^2 = -(dt)^2 + a(t)^2 \left[ \frac{dr^2}{1 - Kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (16)$$

$$\text{or } g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a(t)^2}{1-Kr^2} & 0 & 0 \\ 0 & 0 & a(t)^2 r^2 & 0 \\ 0 & 0 & 0 & a(t)^2 r^2 \sin^2 \theta \end{pmatrix} \quad (17)$$

The factor  $K$  is called the curvature of the universe, and  $K = -1, 0, +1$  correspond to open (hyperbolic), flat (Euclidian), and closed universe (spherical). What is meant by open, closed, flat etc, will be explained later. The time coordinate  $t$  is called the cosmic time.

The reference frame in which the metric has the form (16) is called a *comoving frame*. An observer who is at rest in this frame is called the *comoving observer*. A comoving observer is the only observer that sees the isotropy and homogeneity of the universe (of the CMB). To understand what comoving means, consider the universe to be divided into grid lines. A grid with four vertices is shown in the figure. The coordinates of two vertices at some time are  $(0,0)$  and  $(1,0)$ . It means that the points are separated by one unit of distance. With the expansion of the universe the grids expands. Though the physical distance between the points increases the comoving distance remains one unit. Simply put, comoving distance is the coordinate distance. Observers that are sitting in these grids are comoving observers. It can be also shown that the

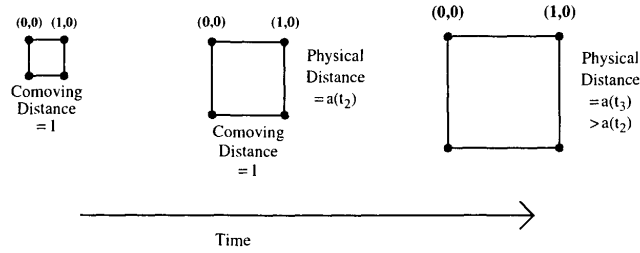


Figure 13: A comoving frame in expanding universe. Figure taken from Dodelson book.

worldlines of particles which are at rest in the coming frame are geodesics. In other words,  $x^i = \text{constant}$  worldlines satisfy the geodesic equation. There are various ways to measure distances in cosmology that will be discussed in a separate chapter.

An observer who is traveling say at a very high speed through the universe can not use (16) to describe the universe. The observer need to Lorentz transform (16) to his/her coordinate.

### 3.7 Cosmological Redshift

In this section we show that red-shift of galaxies that Hubble observed is due to the expansion of the universe. Consider a galaxy whose comoving coordinate is  $(r, 0, 0)$ , and that of an oserver in earth is  $(0, 0, 0)$ . This choice is general – the observer can always rotate its coordinate system to get rid of the  $\theta, \phi$  coordinates. A light ray is emitted by the galaxy and is observed by the observer. As a light rays travels along the null geodesic  $ds^2 = 0$

$$dt^2 = \pm a(t) \frac{dr}{\sqrt{1 - Kr^2}}.$$

We must keep the negative sign because, a light ray starts its journey at  $t = t_e$  from a distance  $r$  and reaches  $r = 0$  at  $t = t_o$  when it is observed. In other words, the negative sign reflects the fact that the distance of an emitted ray to the observer is decreasing. We can then write

$$\int_{t_o}^{t_e} \frac{dt}{a(t)} = - \int_r^0 \frac{dr}{\sqrt{1 - Kr^2}} = \int_0^r \frac{dr}{\sqrt{1 - Kr^2}}.$$

The right-hand side of the equation is a constant, obviously.

Now consider two rays consecutively emitted by the galaxy at times  $t_e$  and  $t_e + \delta t_e$ , and detected by the observer at times  $t_o$  and  $t_o + \delta t_o$ . We can write

$$\int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{t_e + \delta t_e}^{t_o + \delta t_o} \frac{dt}{a(t)} = \int_0^r \frac{dr}{\sqrt{1 - Kr^2}}.$$

From both sides we subtract  $\int_{t_e+\delta t_e}^{t_0} (dt/a(t))$  which gives

$$\int_{t_e}^{t_e+\delta t_e} \frac{dt}{a(t)} = \int_{t_0}^{t_0+\delta t_0} \frac{dt}{a(t)},$$

As the intervals  $\delta t_e$  and  $\delta t_o$  are small, the scale factor  $a(t)$  remains constant in these intervals:  $a(t_e) = a(t_e + \delta t_e)$  and  $a(t_0) = a(t_0 + \delta t_0)$ . So in the above integrations the scale factor can be taken outside the integration which yields

$$\frac{\delta t_e}{a(t_e)} = \frac{\delta t_0}{a(t_0)}. \quad (18)$$

The emitted and detected rays are subsequent. If the emitted and detected rays have frequency and wavelengths as  $\nu_e, \lambda_e$  and  $\nu_0, \lambda_0$ , respectively, then

$$\nu_e = \frac{1}{\delta t_e}, \quad \nu_0 = \frac{1}{\delta t_0}.$$

Substituting in (18) we get

$$\boxed{\frac{\nu_0}{\nu_e} = \frac{a(t_e)}{a(t_0)}, \quad \text{or,} \quad \frac{\lambda_e}{\lambda_0} = \frac{\nu_e}{\nu_0}}. \quad (19)$$

From the definition of redshift we get

$$z = \frac{\lambda_0 - \lambda_e}{\lambda_e} = \frac{\lambda_0}{\lambda_e} - 1 = \frac{a(t_0)}{a(t_e)} - 1,$$

or

$$\boxed{1 + z = \frac{a(t_0)}{a(t_e)}}. \quad (20)$$

If  $z > 0$ , known as redshift,  $a(t_0) > a(t_e)$  which indicates expansion. If  $z < 0$ , called blueshift,  $a(t_0) < a(t_e)$  which indicates contraction.